Fund Flows and Income Risk of Fund Managers

Cen, Dou, Kogan, Wu

Discussant: Philippe van der Beck[1]

[1]Harvard Business School

Background

- Over 80% of US equities managed by institutions (on behalf of households)
  - E.g. \$10 trillion in active mutual funds
- Frictionless view: This delegation is a veil.
  - Managers inherit households preferences.
  - How? Optimal contract: Manager compensation tied to (excess) returns they generate for the investors.
- Reality: Managerial decisions $\neq$ household decisions.
  - Compensation depends on tracking error (Basak and Pavlova [2013] etc.)
  - Compensation depens on fund size/inflows. (Dou et al. [2022]. etc)
- This paper: Instead of speculating about compensation contract, let's look at data! **Awesome!**

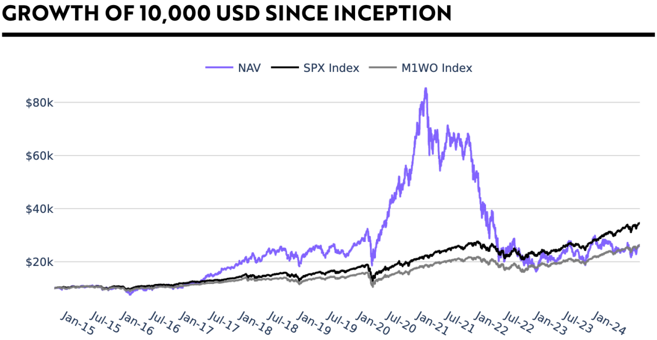
Incentives in Active Management: ARK Innovation ETF

ARK INVEST | ETFs

FUND DETAILS	As of September 30, 2024
Ticker	ARKK
Type	Active Equity ETF
CUSIP	00214Q104
ISIN	US00214Q1040
Primary Exchange	NYSE Arca
Inception Date	2014-10-31
Expense Ratio	0.75%
Fund AUM	$5.78 Billion

- Expense Ratio: 0.75%
- Current Assets: $5.8 Billion (**$28 Billion in 2021)**
- No performance fee.

Incentives in Active Management: ARK Innovation ETF



- Morningstar: *"Cathie Wood's Ark Invest has destroyed $14 billion in wealth over the past decade"*
- ... but it has raked in almost $1 Billion in fees!
  - Cathie Wood's Net Worth 2024: $300 Million (Forbes)

This Paper: Summary

- ARK is an extreme example of (perhaps) wrong incentives in active management: Maximize fund size as opposed to expected returns.
- This paper: Systematic evidence. Construct new amazingly rich dataset on US fund managers' pay, fund size and performance.
  1. Fund returns, flows, expense ratios (CRSP)
  2. Fund managers (Prospectuses, LinkedIn, Websites)
  3. Managers' pay (Census LEHD)
- Main Regression including manager (m) and year (t) fixed effects:

$$\underset{}{ln(Pay_{m,t})} = \underset{[0.13^{***}]}{\gamma} ln(A_{m,t-1}) + \underset{[0.02]}{\gamma^R} R_{m,t-1} + ...$$

- Manager pay driven by fund size as opposed to performance!
- Splitting manager pay shows return performance affects bonuses (inferred) but not base salary.
- Many other cool results. (job turnover, AUM reallocation within fund families etc.)

Comments

- **Great!** Systematic evidence on misaligned incentives as opposed to individual anecdotes (like ARKK).
- Small Comments
  1. Two layers of delegation. Evidence for incentive misalignment at which layer?
  2. Scale versus skill. Instrument based on mandate-driven demand shocks.
  3. (Contribution beyond Ibert et al. [2018] and Naim and Sokolinski [2017])

Comment 1: Two layers of delegation.

- Main result: Managers are rewarded for fund size as opposed to fund performance. Conflict of interest!
  - Prefer stocks that increase inflows over stocks with high expected returns (e.g. ARKK, JanusTwenty Fund)
  - Spend more effort on adv., networking... than predicting returns
- What causes this conflict?
  1. First Layer: Delegated management (+ messy returns)
  2. Second Layer: Specific compensation contract of fund manager.

Comment 1: Two layers of delegation.

- Two Layers of delegation: Which layer's frictions drive results?
  1. **Households delegate management to funds.**

$$\underbrace{Flows_{m,t}}_{\approx \text{Revenue}} = \beta_1 R_{m,t-1} + \underbrace{Marketing, Fees, Networking\ldots}_{\text{Majority of Variation}}$$

    - Small fraction of fund-flows/assets/revenue explained by performance
    - Performance measurement issues $\rightarrow$ Misaligned incentives
    - Maximize fund size $\neq$ Max performance.

  2. **Funds delegate security selection to fund manager.**

$$ln(Pay_{m,t}) = \gamma ln(A_{m,t-1}) + \ldots.$$

    - Incentives seem aligned: Maximize fund size!

- Friction seems to be primarily in layer 1 (delegation + messy returns), not in layer 2 (manager salary).

Comment 1: Two layers of delegation

- Nevertheless flows weakly driven by past returns (flow-performance relationship) and manager compensation is not. How important is this **additional** second layer friction?

- Small fraction of active MFs have performance-fees (e.g. 2:20)
  - In 2000 it was 10% (Elton et al. [2003]). What about today?
  - Relationship between performance and manager pay should be **stronger** for funds with higher performance fees. You can test this!
  - If not, additional friction in Layer 2![2]

[2] Or (less likely): results are driven by measurement of performance, difficulty of separating scale and skill, econometric specification etc.

Comment 2: Pay for skill or AUM

- Causality: Is correlation between pay and AUM evidence that AUM-pay (as opposed to skill-pay) is contractually specified?

  (Which would mean strongly misaligned incentives)
- What would Berk and Green [2004] say?
  "*You need an instrument for AUM*"!
  - Fund size (not fund performance) is the right measure of skill.
  - Therefore we expect a significant loading of pay on AUM
  - Raw AUM doesnt provide evidence for friction in managerial pay.
- Realized returns are noisy! Unlikely that households can perfectly infer manager's skill from past performance.
  - Controlling for past returns, size probably not a good measure of stock-picking skill. (Song [2020])
- Personal opinion: Manager fixed effects (θ_m) somewhat sufficient.
  - Why would managerial skill/their stock-picking ability change over time?

Comment 2: Pay for skill or AUM

- Say you really want an instrument for AUM (because referee or academic body thinks that manager fixed effects not enough)

Exogenous variation in AUM unrelated to managerial skill

- Current instrument: Changes in AUM due to **non-fundamental** returns (supposed to capture "Luck")
  - Stock returns driven by institiutional demand shocks: Inclusions in investors' mandates. It is precisely a skill to predict those!
  - Countless arbitrage strategies built around front-running index reconstitutions, flow-driven trades etc.
  - Generally: It is a managerial skill to pick stocks which will be in high demand in the future (regardless of whether this demand is "fundamental" or not)
- Alternative Instruments: Changes in AUM based on
  1. Morningstar Ratings Changes See e.g. Ben-David et al. [2022a] Ben-David et al. [2022b] etc.
  2. Fund name changes See e.g. Cooper et al. [2005]

Other Comments

- Relation to Ibert et al. [2018]: Same results for Swedish data.[3]
  - *"While Ibert et al. (2018)'s results are intriguing, it remains to be seen how applicable they are to the larger and differently-regulated U.S. mutual fund industry."*
  - What regulation would make you think their results wouldn't hold in the US?
- Is there a better motivation?
  1. Important next step is to link manager pay and stock selection (holdings data). Clean holdings data available in the US.
  2. Much greater fund universe in US, so more power in testing further splits: E.g. Stronger pay-performance relationship for funds with performance fees.
  3. Additional split by bonus pay ✓. Nice! But results probably strongly depend on definition of bonus. What happends when you vary the 20% threshold?

[3] Also: Naim and Sokolinski [2017]!.

Other Comments

- Is this evidence for flow hedging? Managers tilt towards low flow-beta stocks to hedge income risk (Dou et al. [2022])
  - Would rather expect the opposite: Managers tilt towards stocks that attract a lot of flows (trendy sentiment stocks)
  - These stocks likely have high flow-beta?
- Main Body: 60 pages. No Appendix.
  - Would be very helpful (for the reader) if you distill the main results and allocate data construction, econometric details etc. to the Appendix.

Exciting Next Step: Linking Manager Pay and Portfolio Choice

Ticker	Name	Market Value	Weight (%)
TSLA	TESLA INC	$901,385,209.17	14.21%
ROKU	ROKU INC	$600,635,984.76	9.47%
COIN	COINBASE GLOBAL INC -CLASS A	$564,017,266.54	8.89%
RBLX	ROBLOX CORP -CLASS A	$427,197,773.51	6.73%

- Fund-size more important than expected returns. Incentive to tilt towards hype stocks. Fee-maximizing portfolio $\neq \Sigma^{-1}\mu$
- Asset pricing implications of these misaligned incentives potentially very large! (At least for mispricing in individual stocks)

Summary

- Intruiging, fun and important paper!
  - Systematic evidence on misaligned incentives as opposed to individual anecdotes.
- Small comments:
  - Frictions in both layers of delegation. Currently evidence for mix of frictions in both layers. Can you disentangle further?
  - Do you really need the section on causation?
    Page 56-64 in **Main Body?**

Literature

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